ActInf Livestream #009.0: “The Projective Consciousness Model and Phenomenal Selfhood" (2018)

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Duration: 1:48:16 | **Uploaded**: Unknown **Transcript method**: auto_caption

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hello and welcome everyone to the active inference live stream I believe we are live it is ACM stream 9.0 and it is November 18th 2020 I am Daniel fredman and I will be doing a solo active stream today so welcome to teamc com everyone we are an experiment in online team communication learning and practice related to active inference you can find us on Twitter at inference active active inference gmail.com our public keybase team at our YouTube channel or um through any other mechanism this is a recorded and an archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here and for video etiquette on live streams make sure to mute if there's noise in the background raise your hand so that we can hear from everyone on the stack and use respectful speech Behavior so just a quick announcement before we jump into the paper discussion you can find on our Twitter information about the rest of our discussions for 2020 we have six more uh group act imp streams in 2020 on November 24th and on December 1st we're going to be in act imp 9 discussing the projective Consciousness model and phenomenal selfhood and that will be the paper that this actm stream 9.0 is context for in active 10 on December 8th and 15th we'll talk about the paper a variational approach to scripts while in actm 11 on sophisticated affective inference we'll talk about more modeling and math again and that will be on December 22nd and 29th so that will close out 2020 and then we look forward to a whole new panel of events and projects collaborators for 2021 so find out more on our Twitter that way let's talk about what we're going to do in actim stream 9.0 which is right now so the goal of actim 9.0 is going to be to set the context for 9.1 and 9.2 which are going to be the group discussions that we have on these papers and the paper that we're going to be discussing for 9.1 and 9.2 is called the projective Consciousness model and phenomenal selfhood by Will aord at all 2018 and there's a link that people can see the video that we're about to jump into right now is an introduction to the context of some of the ideas presented by the authors it's not a review or a final word it will contextualize some of the Ideas Math and vocabulary that might be interesting or important to know for reading will afford at all 2018 and the punchline that they're going to work their way towards is that we might be able to combine Fields such as geometry cybernetics and neurophenomenology using the framework of active inference and the free energy principle to potentially understand or explain Consciousness so that's going to be up to you up to us how we feel about these kinds of claims that they're going to get to in the paper but that's where we're headed and the sections of 9.0 will be as follows first we'll go through the keywords and hopefully provide some background on the different keywords whether it's your first time hearing one of these words or whether you're in one of these fields and then we'll talk about the aims and claims of the paper go through their abstract and the road map the sections then we'll talk a little bit about Consciousness from the author's perspective just what do the authors say is important about Consciousness and what do they think are invariant aspects of it we'll then walk briefly through the figures just try to understand what are these figures even doing how are we going to get a first pass on understanding what these figures are all about and then in 9.1 and in 9.2 we're going to stay with this exact same paper so save and submit your questions and feel free to get in touch with us if you'd like to participate whether that's in 9.1.2 or in a future Endeavor let's talk about the keywords the first keyword is uh actually not on the paper keywords but is what we're about here is active inference and the free energy principle that's what we'll try to tie so much of this back to and here where the keywords that were listed cybernetics projective geometry Consciousness neurophenomenology firsters perspective and perspectival Imagination so cool terms maybe it's your first time thinking about math and geometry in the context of Consciousness maybe it's not but hopefully as I go through these following keywords I hope that this is interesting and exciting whatever level of familiarity you have with these Concepts so first keyword is cybernetics and it's also written there in Russian as kuberi and um there's many resources to read on for the history of cybernetics some of which we're going to talk about here because in a lot of other act dstreams and other locations you can probably hear about the mathematics underlying cybernetics and about how it's related to control both things that we've talked about on this stream but let's talk a little bit about just more historically or from a um cultural perspective what is cybernetics what is it trying to do cybernetics is a school of thought tradition of action however you want to think about it that's related to the action policy selection and control of goal oriented systems now there's various orders of cybernetics which we can sort of walk through here and um I'll go through a couple of different resources on cybernetics and hopefully you'll see soon why I put a history citation on the bottom left and a little bit of Russian on the top and it's not just because we have great Russian colleagues in team comp here are the orders of cybernetics there's probably different ways to State them or think about them first order cybernetics might be thought of as self-regulation without detailed observations so this could be like a negative feedback system that is able to maintain a semblance of homeostasis in virtue of being an isolated or a closed process like there's something that gets triggered when it's too hot and then as soon as it's triggered that it's too hot it instantly kicks in a cooling program second order cybernetics is um self-regulation that can be influenced by observers within a domain of control so that could be for example a continuously varied um thermometer that's able to enact different kinds of control policies as a function of continuous inputs from the thermometer third order cybernetics is self-regulation that is is influenced by observation with a relational basis or an ecological grounding so the system in this third order of cybernetic Loop is able to acknowledge its surroundings and learn develop and evolve and fourth order cybernetics is when self-regulation involves the enactment of multiple different realities or adjacencies affordances counterfactuals a lot of ways you can think about that in uh terms of a functioning creative Observer so these are deeply contextual utiliz and Integrated Systems that are doing higher orders of cybernetics and I'm not a stickler about which order is which number I also on purpose drew the parallel with the um one two and three Loop thinking which is related to um some decision-making Theory from Innovation and the entrepreneurship space but I just thought it's interesting here's a figure from the um book chapter cybernetics in the 20th century on the bottom left there and it shows a map of Concepts and how the related to cybernetics so on the left are sort of two inputs to cybernetics which are control theory and the idea of control and then communication Theory and the idea of communication and information and so those are two branches that we've talked a lot about on this stream and we will continue to talking be talking a lot about because control of action and communication message passing these kinds of things inference is the thread that runs through them all these kinds of things are critical for control theory for cybernetics for the kinds of systems we want to get into working with on the right side we see cybernetics branching back out into the machine area with the engineering sciences and the planning of for example complicated Factory operations but as well as into the animal and the social levels so the animal cybernetics is about for example the self-regulation of behavior and the social cybernetics could be about um the way that our governance is run and there's a really interesting book read plenty that's kind of like historical fiction about cybernetics speaking of History this is such an interesting plot and it was in the cybernetics book figure six it is the usage of the term cybernetics and control by years in the text of populations Publications indexed by Google Scholar so caveats maybe people are speaking about it with different words or it's being used uh different language not English or there's some other way of saying it but look at how similar these things are in terms of their Trends where both of them ramp up gradually leading to in early 2000s a lot of t discussion of cybernetics and of control potentially looking backwards a couple years so think about these peaks in the early 2000s as potentially reflecting something at the close of the 1900s and then how they both start to drop off and this is consistent with something that a lot of us have experienced which is that cybernetics is a term that a lot of at least English speakers may not be too familiar with but we can look at the journals that friston and others have published in we find a lot of connections to cybernetics what is going on there so to get a little sense of what is going on two other resources to look into would be this big history understanding of complexity information and cybernetics and then this book which I'll read a quote from right now which is called from new speak to cyber speak a history of Soviet cybernetics by Slava gich and the author writes in this book from 2002 the history of Soviet cybernetics followed a curious Arc in the 1950s it was labeled a reactionary pseudo science and a weapon of imperialist ideology with the arrival of kev's political thaw however it was seen as an innocent victim of political oppression and it evolved from a movement for radical reform of the stalinist system of science pretty interesting in the early 1960s it was hailed as a science in service it was hailed as a science in service of Communism but by the end of the decade it had turned into a shallow fashionable Trend using extensive new archival materials gerovich argues that these fluctuating attitudes reflected profound changes in scientific language and research methodology across disciplines in power relationships within the scientific community and in the political role of scientists and engineers in Soviet Society his detailed analysis of scientific discourse shows how the new speak of the late stalinist period and the Cyber speak that challenged it eventually blended into cyber new speak so maybe there was a a book summary of this um work but pretty interesting how that plays out and so definitely we'd love to learn more about this whether uh we have Russian speaking colleagues who could fill us in or anyone else next keep word projective geometry what is projective geometry about well of course you could go down the rabbit hole with this topic but let's just try to keep it constrained to what the authors are going to use it for in terms of the paper they write the concept of a 4D four-dimensional projective transformation is Central to the model as it yields an account of the link between perception imagination and multipoint of view action planning great okay so what is happening here and how can we understand this and its relationship to projective geometry I'm going to introduce just two ideas here with this visualization the first by looking at the bottom is the distinction between the ukian and the non- ukian types of spaces and then the second idea with a purple arrow is this idea of a map or potentially you could call it a mapping because it's actually a process and it's actually a certain type of process it's a projection so when we're talking about projective geometry we're talking about geometry which means the The Shape of Things and the spaces that they're existing in so as opposed to topology which is how things are conned connected with nodes and edges geometry is about things and their shapes and their spaces we're going to talk about projections amongst different geometries specifically potentially this one between ukian and non- ukian geometries and here's a quote that they write authors do they write this perspectival phenomenal space so the space in which they're going to say that experience occurs within phenomenology occurs within a space allows us to locate ourselves and navigate in an ambient space that is nevertheless essentially ukian however in a ukian space there are no privilege points no points at Infinity no Horizon and no and an arbitrary origin they say there's no nonarbitrary origin same thing this very this very strongly suggests that a non- ukian frame must be operating in the phenomenal space with the help of sense motor calibrations to preserve the sense of the invariance of objects across the multiple perspectives adopted and relate the situated organism to the ambient ukian space so what is it that allows me to continue thinking about this as a pen even as it comes very close to my face and looks very weird so what is that generative model that allows me to act like everything's normal as objects maintain their invariances as the eye travels through the space and so we can contrast this ukian space that they're supposing that we exist in they're saying we exist in a space that doesn't have privileged points of access no points at Infinity etc etc um and they're saying the space that an organism for example moves through is ukan while the space that their Consciousness is being projected into experientially is non- ukian what about it is non- ukian well let's just take the opposite of what's in red and put it in blue so there are privileged points from the point of view of the Observer that is the privilege point of view quite literally there can be points at Infinity there can be points that sort of just like they're so far away they're infinitely far away and not just from an affordance point of view but from a perception point of view there can be a horizon in fact we see it every day and there can also be a non-arbitrary origin which is a little bit of a restatement the idea that there could be a privileged point but basically it's privileged it's a non-arbitrary origin these are pretty similar Concepts now non- ukian it's like saying nonlinear and the joke about this is things that are not elephants that's the non- elephant class you'll know it when you see it so that's about how helpful it is to say it's nonukan or it's nonlinear model okay we get it it's not linear model it's not ukian geometry but there's many choices and the reason why even though it sounds like wait you're barely ruling anything out we're actually ruling out very specific parts of the geometry of the ukian space so in the ukian space parallel lines don't intersect but that's not necessarily um the case for certain types of geometry or a triangle with three edges and three angles is 180° from the sum of the angles but on a sphere it's not that way so that is what is happening with these geometric projections is we're going to be projecting from sort of the um grid they want to think about now you might want to ask Bucky Fuller if we're actually in a ukian space or if it just seems that way because of public education but for the purposes of this paper we're in a ukian space as action and that's the space we want to be making our perception uh simulate so to speak and to act like everything is normal and to decrease our uncertainty about a ukian mapped space while also relating it to through a projection something that does have a privileged point of access and something that is as they put it informed by sensory motor calibration and relates to our action affordances so that's how you can have a ukian concept of for example um a gap in the ground that you can't jump over so it's a very complex estimation of what you can do and how big the Gap is and where you are how much runup you would need Etc all of that is going to come together by doing what they're going to talk about here now Consciousness um it's great to talk about Consciousness and I'm going to go into a little bit of detail soon about some of my own thoughts on the issue but when talking about Consciousness it's important to be really clear even though it's something that we're not always sure about It's always important to at least use language very carefully and use our scholarship and our Intercultural respectfulness very importantly because as I'll hopefully come back to a few points soon it's just a critical issue and we can't think of it um just flippantly the authors write when talking about Consciousness because I want to just go with what they think about Consciousness for here they right we might divide contemporary neuroscientific and psychological theories of Consciousness into three main types the biofunctional the cognitive neurofunctional and the information theoretic as should be clear the pro Ive Consciousness model the PCM incorporates the emphasis on self-awareness common to many approaches to Consciousness as well as an emphasis on the biological functions of Consciousness so they're defining the space a certain way I'm going to Define it a different way soon but they're defining it that way then we're going to insert the

PCM here and then they write at the end the PCM thus subsumes and unifies the main insights in all of these approaches to Consciousness above all what we add to pre-existing Theory is a geometry the thesis that projective Transformations and projective frames necessarily subtend the appearance and workings of Consciousness so whatever we're going to learn in this ACM stream or whatever we think the PCM is doing and we want to evaluate it not by um what it doesn't do but what it does do and what it claims to do what it sets out to do that's where they're trying to step in their understanding of literature and Consciousness is that they broke Consciousness down into these three kinds which ation exist there you could read the paper and then they're going to step into that Gap and then they're going to make the claim that they're subsuming and unifying the main insights from these approaches where does that leave these approaches from a theory perspective question for another day but that is where the authors understand their work to be fitting in so if you don't know what the PCM is or it's kind of trippy to think about what it means to do mathematics and geometry of Consciousness that's perfect here is the blank that they're going to fill in with their work what what do I think about Consciousness though and I want to be clear about what my stance is or at least just give the tip of the iceberg on my stance because this is a first person actim stream and uh there's some relevant research I'll discuss in a few minutes and also I've just for a long time been interested in the science of Consciousness but also this second layer of How It's communicated especially from scientists so there's a lot of places to go here and a lot of places to start I'm just going to walk through a few General thoughts and raise a few questions questions really more questions than answers and then I'll discuss a paper from the last couple of years um with a colleague Eric sovic all right so there are many ways to think about studying Consciousness and one path that you could go down um this is sort of you know across cultures through the time One path you can go down is thought experimentation so this could look like meditation it could look like a specific thought experiment um or some other sort of insight approach or a gankin approach and whether it's culturally called one thing or another is not as much my concern here um the thought experiment approach doesn't involve making interventions for example in the outside world just thinking about Consciousness you know hearing TED talks and stuff like that so the issue is that um whatever is actually happening if that's what we're pursuing um we could just be led to the wrong conclusion and then if there is some element of free will or interventionary role of Consciousness then we could intervene to um end up believing that couldn't intervene or any of these combinations so I just never quite understand the relationship between what one directly feels convinced about with respect to a thought experiment or a thought or an idea and how that might relate to how it actually is for people who aren't that person and one aspect of this is that it's really impossible to know which aspects of thought experiments can be transferred to other situations so for example one of the most famous papers in this area most cited papers is a 1974 paper by a philosopher named Nagel and that raises the question what is it like to be a bat I don't think he invented that question um but he's the one who got famous for it and in that paper it's written an organism has conscious mental States if and only if there is something that it is like to be that organism something it is like for the organism now you might think what are philosophers doing because it sounds like a sort of a weird question and I understand that many uh careers and a lot of good thinking has come out of this question so I want to optimistically see it as a catalyst in a seed um it's always good to have simple questions to return to it's good that it's such a simple question and it opens to so many perspectives it initiates students into our knowledge Traditions there's so many advantages to having these kinds of classic works but to be that um uh to be blunt I guess I found the phrasing of this question of Consciousness to be quite unhelpful because it is so informal it's so hypothetical it's untestable it's ambiguous but it's specific it brings a bat in to question and then you can imagine oh is a dog what is it like to be a dog now these are all good things to imagine but again they don't really bring us closer to addressing real Frameworks for Consciousness and the reality is there's been now a vast Wasteland of literature that's just not comparable to other work in the same exact space across different species that essentially amounts to speculation and appealing to what it might be like to be a certain type of animal thinking well if I was really a bat and I grew up as a bat would I ever like thought uh what it was like to be a human how I so it ties up to all these second level questions that aren't super relevant and um we've always wondered through time every culture everyone has wondered what other systems experience whether that's a person or an animal or a fictional character or a physical object so the speculation and the think pieces they don't really add something substantial to the literature which is what the scientific literature is about for me so the bigger issue though with thought experiments of any kind is that we can always imagine things that are just not making sense Within framework or any framework or just don't relate to the world that we live in um we all know that we can be wrong about an idea but believe that we're right and so um to kind of rephrase that in an active inference framework we could say that these are situations where the Precision parameter is very high there's a belief a metacognitive belief that the Precision is high that some question has been resolved um this represents that um for example someone's model of Consciousness whether it came from a a knowledge tradition or their own thinking they believe they're they're onto it they got it figured out it's resolved for them but um that doesn't mean that your s your estimate is related to S hat the actual state in the world and I'm not passing judgment on being overconfident or um being wrong because sometimes it can be very important and Powerful to have these kinds of beliefs but that's not what being correct is about and so if you actually don't care about being correct being right going towards the truth and you really don't care about that then um I'd love to hear where you're coming from um the rest of the scientific literature although with many shades and many differences is moving towards what actually would be General generalizable through truth so let's go back to consider experimental approaches which is something that comes up a ton in the philosophy and the science of Consciousness literature now it would be awesome to do experiments on Consciousness and in some way don't we all we could study humans as they behave and sleep and daydream take um psychedelics or they wake up for anesthesia and they develop and there's so many things that could be exciting to learn about also if we can do experiments on Consciousness we could do experiments on animals now we could do experiments that are interventive like um manipulating the brain of an animal or the body or the situation of an animal these experiments might be seen as inhumane or just to massively violate ethical standards for humans and so there's a bit of irony in doing these potentially invasive non-consensual experiments with captive or wild animals when we're studying potentially their capacity to not consent to our discussions with them um it's a interesting question I won't go down the ethics uh route here but a non-ethical question that's related to experimentation is that we just don't know which systems to study or what observables to measure for experiments and so we're studying something that isn't really measurable or observable but we'll come back to this idea of observability in a few minutes when we're talking about neurophenomenology and the reality is no matter what you do measure even if you come up with this one special measurement that hasn't been done before all experiments are interpreted within a cultural and a scientific milu in terms of what measurements conclusions interpretations are ethical correct accepted valid discussed all these things and even going another layer back about experiments even assuming that we can measure what we want to know and measure and everything experiments um especially if they're disconnected from grounding in theory it's difficult to know what is really being uniquely tested or demonstrated or falsified by different kinds of experiments or what we can learn from different kinds of measurements and um there's a lot of ambiguity a lot of levels the ambiguity can enter into but just to give a few examples an experimental result could be consistent with many different Frameworks or potentially all Frameworks or a set of Frameworks for Consciousness so when we really want to be identifying what Frameworks we're wanting to work with it doesn't narrow the space as much to do experiments which will return too soon and that's kind of related like the neural correlates of Consciousness debate which is do you put somebody in a neuro imager and then have them do a no report Paradigm where they don't say yep I heard the sound so you play sounds of different uh you know notes but what if it just subconsciously influence their brain there still could be a neural signature of that how would you know they experienced it well that's called a report Paradigm you ask them did you experience it but in some sense you're just asking them if they remember or if they hit the button so it gets messy but that really gets into some of the the Weeds about what you're measuring and another example or area where we see limitations on experiments related to Consciousness is an animal behavior so animal behavior seems naturally like it'd be adjacent to human consciousness studies and we see a wide variety of things being used here like pulley tests can two of the same animal pull a pulley so they get a reward looking in a mirror can one recognize themsel and take an action pattern to clean thems in the mirror and the first thing to note is that whatever kind of setup you're dealing with a visual illusion a cooperation task a recognition task memory planning they're always going to be based upon the affordances of that animal and again it might be ambiguous as to whether they're using some CU that you actually expect um or if it doesn't see it's not going to be able to be in the mirror test so at the very least these tests for Consciousness which we'll get back to don't help us really study the ground rock of this Frameworks for Consciousness question because all you could say is okay this bird can do the pulley test but we don't even know what that means and then of course if we find out that potentially experience isn't related as much as we thought to behavior or one has a model that it's not related to behavior then you can say yeah it's a robot that just acts like it's a crow all right I think I'm still streaming but it looks like my camera froze I'm going to just replug it in um yeah actually I'll just see if don't know what it's about but that happened uh last time as well but sayy I'll just close the camera for now um actually let me just see if I can remove the source and then add the camera back in we're going to find out live whether it is OBS being weird or whether it is um the actual USB connection and turns out it was OBS we now know and we're right back in it so um to the point about potentially robots or zombies as denet might put it doing uh complex behavioral tasks uh you wouldn't know whether the experience was being had even if the behavioral experiment you had lined up for this animal was enormously complex and uh robots might be able to do things that we might not expect them to be able to do already such as uh learning language or being culturally embedded and so yeah another Point here is that before you get too excited about any single experiment that might prove or disprove a model of Consciousness consider it just another grain of sand on a sand Heap and any single grain of sand that you're adding to this Heap now could be the one that sets the whole Cascade in motion so by all means I encourage experimentation but that being said let's not fool ourselves in pretending that many of the data points we already have are out there in the world already and that just just one more experiment is unlikely to provide any type of knockout evidence for or against a model of Consciousness and so to kind of um recap there there's the thought experiment Branch the philosophy meditation Insight route and there's the experimentation interpretation route and uh those are connected of course experiments are ideas too but that being said they both have pros and cons and people will often argue one or the other as if they have um the answers from one branch or the other I hope I've demonstrated that that isn't true and one reason I think that it's important to have this Viewpoint is that we're living in a world just as a prima fascia observation where there's some people humans who say that they think all things are Consciousness or that all things are Consciousness in their own special way or that nothing is conscious including humans and of course there's many flavors to each of these viewpoints that could be held um or at least communicated and there's many things about beliefs about Consciousness that might not be verbal or expressible in a certain language so the point is we're in a total state where the evidence in the world has resulted in people believing different things through time different things even within the same family and so let's be humble the table has mostly been set for us in terms of observations about ourselves and about our world though it doesn't mean we can't reimagine much of our own experience or the world but um it would really have to be quite the thought experiment or quite the experiment uh in the real world if it can resolve these long-standing and philosophically grounded differences in worldview between schools of thought and between people and even if it did this even if the experiment were so good that it just convinced everyone it wouldn't even mean it was right it would just mean that it was convincing as a meme and so if we're really trying to find out what's right rather than how can we shape our own reality and other peoples which isn't always the worst task if we want to be right then as far as I know it's not going to be the experiment or the thought experiment route alone and one last point on this slide because there's a lot to say here is that there's academics who believe different things and you can see these slides and citations and see that um they cite different work that says one thing about the global workspace or another thing about information Theory there's also different views on Consciousness from different world Traditions from artists from practitioners from non-scientists from non-philosophers from all kinds of people people children adults everyone this is just to say that we should be very cautious about any kind of hegemonic especially scientific effort to define consciousness whether it comes from Academia or from some other source there's just way too much on the table here for people to be making sweeping claims about what is or is not conscious uh and there's a lot that is Downstream to that it's really a personal and a cultural thing what somebody believes about Consciousness um what they want to say or what they want to commit to and um also while some people might find these topics exciting I think they're exciting if you're listening you probably find them to some extent exciting as well but many people are not interested in considering it and don't perhaps want to be told this is how their Consciousness or experience is and other people are simply okay not knowing whether they're curious about it or not and so I'm always skeptical about science says type thinking which can be often how scientific papers are written even if it's written in an author's say type thinking it's about how it's communicated we've all seen the misuse of science A's policy related to everything and uh Consciousness would be the next level on that wouldn't it be so fill in the blanks think about what kind of power is held by the people who can say what or who is consciousness or evaluate conscious states such as suffering or pleasure so there's just a lot to it and then um the last Point really here is that people can change their beliefs during their own Lifetime and go deeper and shallower into different ideas and that probably doesn't change the underlying nature of Consciousness so it will never be enough how we think at one moment or for a group of people to believe something I've gone down these discussions and Roads many times with people um alone and in conversation and people will tell me well you know we're working with our best possible understanding of Consciousness given the research at this time and it's just simply not true there's so much to know and respect in terms of the research and also the non-scientific literature on Consciousness that there is just a lot more to the discussion than well let's just tell the kids something simple to it now to go beyond that and thanks for bearing with me it was just I wanted to say a few of those things because a lot of times Consciousness papers are approached from non-scientists with excitement so I want to provide a little bit more of a nuanced scientists take here on how we can really respectfully study Consciousness because this isn't just like frisen at all figuring it out in a paper but I love the paper and it's interesting so how do we hold this yes and perspective through Nuance hearing multiple people's perspective having participation so what I'm going to talk about here is this paper that I worked on with my colleague Eric sovic who's a professor in Norway and the paper called the ant colony test as a the ant colony as a test for scientific theories of Consciousness so this is how I came to be thinking about a lot of these topics now Professor sik's background was in neurobehavior and in genetics of the honeybee mine was in similar areas but related to ants Eric is also a very pragmatic and a philosophical thinker his PhD adviser who is Andy Baron uh at M University in Australia is a honeybee biologist and Professor Baron has also written philosophy papers that are related to entomology as well as to experience and Consciousness uh on my side my PhD adviser Professor Deborah Gordon at Stanford is an ant behavioral ecologist who also wrote a lot of philosophy Works broadly construed and so both Eric and I had really positive mentors and role models for how we could combine entomology with philosophy so that's where we were coming from with this paper was how could we combine our Direct ethological behavioral ecological experience with the social insects with what we knew about neurogenetics and molecular biology and then tie that through respectful engagement with some of the philosophy of Consciousness literature here's what we didn't do in the paper what we didn't do is make a claim or a bet on whether ant colonies are conscious or not whether ant workers are conscious or not if people have feelings on that let's hear about them on the actm stream let's find antstream let's do something fun let's hear about it but that's actually not what we did in this paper we do a few things in this paper the first thing is that we proposed this terminology of a forward and a reverse test for Consciousness which I'm going to return to in a second with this image the second thing we did was to review the literature on Consciousness from a scientific literature and make classification schemes for different kinds of theories of Consciousness so you can see why I was interested in the breakdown of this paper's uh understanding about the fields of different scientific studies of Consciousness and then the third part of the paper we used empirical findings from entomology so observations of morphology or of behavior of individuals or of colonies from entomology and we asked whether different Frameworks for Consciousness that we found during the literature review might come to the conclusion that ant colonies Andor workers had Consciousness or awareness and we as authors remained neutral on the issue to the extent we could as far as just presenting hypothetical evaluations of colony Consciousness from the point of view of these different Frameworks which is something that's so intrinsic to Consciousness so it's kind of fun and it turns out that Frameworks just simply wildly diverge in their claims about ant colony Consciousness and it's really not a surprise that it turned out to be that way because the same Frameworks uh diverge in their claims pretty plainly and people diverge in their opinions and so if the Frameworks are going to disagree about humans being conscious or electrons then they're going to disagree about ants and um it was just interesting to experience this though and ask how we could have a reconsideration of what we're really doing here with the Consciousness studies uh area because there's thousands of speculative papers about Consciousness in animals but are we moving the needle and if we are in what direction is that where we want to be moving it what did we do in the paper so we introduced this idea of a forward and a reverse test for Consciousness and we thought thought about forward in Reverse just from the point of view of the Observer the experimenter so that's very related to this phenomenological invariance idea that the PCM model is going to return to in a little bit and this idea that everything in Consciousness should be relational let's take this perspectival and this relational approach very seriously so what is a forward and reverse test a forward test is when you trust your test and it describes the mode of usage of tool where you're measuring systems and drawing conclusions so for for example in the figure it says I trust my test how heavy are these boxes and we have this gray test it says it is an unknown whether the box is conscious or not we put it on there it says yep one and then we go out in the world and we investigate whether different boxes are um different weights or have Consciousness or whatever and that is for validated calibrated instrumentation and it's important for the normal in a paradigm uh vocabulary the normal science which is advancing science and a reverse test is a different phase a reverse test is when you don't trust your test yet and this is like calibration and calibration is when you take things that you do know about like we know that the meter is this much um relative to um this and we know that the wavelength of this is going to help us calibrate this tool um but really just when you use a scientific instrument it has to be calibrated and if we want to have a science of Consciousness we have to have a calibrated instrument and so the reverse test is when you don't trust a test and you don't know what the machine is going to read out but you put on the one PB block and you say I'm going to fine-tune this machine so that it comes out with one when I put the one on there and then we'll know that this scale or this spectr photometer works and it's kind of this back and forth this two-stroke engine that we've seen so much about and we've been talking about um the back and forth between um action and inference of course we've been talking about that from the framework of computationalism from the basian perspective the basian structuralist perspective inative ISM and ecological psychology and just cognition in general agency how do we do this forward and backwards of more tool likee based upon affordances and more inference based upon observations okay so it's kind of a two-stroke engine for studying Consciousness from a field perspective so this is like what should the field of Consciousness researchers be studying now wouldn't it be great if we had a forward test for Consciousness if we had a scale that we could go yep we hooked up this person's brain or this wi-fi router or this Beetle and yeah it's a two out of four it's a 2.9 out of you know whatever it is yeah sounds great I understand why people want to mathematize and why they want to measure Consciousness but if you meet somebody who has a good test for Consciousness then you know get the Tesla Tech get the perpetual motion machine get everything you can from that person because we don't see it within the current space we're in in um we want to have this test prefrontal cortex language whatever it happens to be no caveats just a test of course it doesn't exist and so that's kind of what this issue is about and so when we don't have a forward test for Consciousness even though people might think they locally have one I hope I've sort of dispelled that with a previous slide if we don't have a forward test to operate in the action mode do we have a reverse test to help us work in the inference mode so that is um where professor soic and I wanted to think about different kinds of forward tests for Consciousness and then how could the ant colony be reversed test for Consciousness so here's the framework that we Loosely organized into for our literature framework um in terms of forward tests for Consciousness so this is like what other people might call Frameworks for Consciousness we said look all these models what they have in common is that there's some measurement some tool like intervention or um intervention into the process or measurement that could be done that would help us have a forward test so like is this system conscious does it have X does it have language does it have a prefrontal cortex okay now the reverse test for Consciousness would be like we know humans are conscious so any theory that says that is fine with me or we know humans are not conscious so I rule out any theory that says that right whichever one you think you believe fine so we separated into a few different categories the first one was neurobiological so remember that they had three categories biofunctional cognitive neurofunctional and information theoretic what we did um with our slightly different breakdown of the uh categories in the space not a final word this is just what we had uh found out about is we divided neurobiological models into the following categories which were uh structural models that were based upon macro or micro uh structure so macro structure would be like the cerebral cortex is this size or it has this brain region so gross anatomy macro structure the micro structure is like wow there are so many connections on the dendrites or this cell type has this shape so things that you need a microscope to see macro macro to micro so you need the microscope for the micro histology or oh the architecture of these neural networks is a feedback that means that it could be implementing something so macro micro structure of the brain neurobiology or of the brain in the body for that matter because it's not always about the brain functionalist models for um Consciousness would be like there is a function that does egocentric navigation in crawfish and crows and humans so these are conscious entities so because there's a functional element of something that's being done even if it's being done in non-homologous brain regions or two different ways by two different species or in somebody who is born with vision or who wasn't born with vision it's the function that matters and then another neurobiological assessment that people often make in terms of forward testing is dynamical featur features of brain function so they'll say well because the brain is a complex adaptive system with winterless competitions and leaky integration and action policy selection and um lock of volaric um you know uh Predator prey type Dynamics or chaos and landslides catastrophes power tail of noise all of these dynamical features coming from usually a complexity or a signal processing perspective so those are kinds of appeals that people make to the brain and the body there's also the egocentric and relational Frameworks for thinking about what counts as a model of Consciousness and in this category there's a bunch that we could go into but just to make it kind of clear what I mean by egocentric and relational which is actually the first invariant principle of the PCM so totally agree it's critical it's um the idea of spatial awareness and perception which is um whether it's the lived experience or just the acting as if of things being closer to you being more accessible but everything in conscious is experienced in relation to something that's agreed upon even when people don't even agree what that relationship is or what that something is so that's kind of cool um then there are cognitive and behavioral Frameworks so these are cognitive being broadly understood as like things that computers might be able to do or computer networks or extended cognition embedded enacted encultured all this stuff that we talk about every week let's take that broadly to cognitive so here by cognitive I don't just mean what what people think of happening between you know the student's eye and them filling in the multiple choice test that's not just cognition we want to have a broad understanding of cognition multiscale message passing basian Nets Etc and the categories here are um there's multiple but examples are like emotional or affective States kind of foreshadowing um the sophisticated aec of inference which will be Act 11 but emotional states like regret um or optimism bias and cognitive biases more General so everything from uh visual Illusions to other types of illusions that are not visual but can be related to the order of things being presented these are sort of cognitive effects that potentially like a neural network might fall victim to and so what we've experienced is that there's this runaround and people will say well this bird has this brain region so it's a macr structural claim so it's conscious and then somebody brings up a a shortcoming of the idea of macr structuralism like well does it really mean that it's aware just cuz it has this brain region what about sleeping people or what about you know Daniel dennet or something like that and they go well right it's actually the brain region and the micro structure okay so it's micr structural great well then would anything with that micro structure have that kind of a characteristic um like if you made a model in a computer we did have awareness well no it has to be evolved or it has to have language or it has to be implementing something ecologically great so we want to chase that rabbit down because we're serious about answers here and so we want to chase it down someone says oh well when I said language I didn't just mean a chatbot I meant they they they feel emotion you say well but what if the chatbot says it feels emotion too what are you going to do and then all of a sudden that whole feeling emotion um communication of a feeling emotion all of a sudden they have a new explanation so let's keep on chasing down these rabbit holes and uh one other category that is closest to what is described in this paper is the mathematical framework so we took a step beyond the authors in the um 2018 paper the projective Consciousness model where they said there's the information Theory category of Consciousness models we actually separated a few of these models of mathematical Frameworks so we had decision- making and long-term planning um as well as information the theoretic and GE geometrical so there have been some uh maybe it was published similar or slightly after this paper but there have been some uh geometric Theory Of Consciousness info uh Theory and then also a lot of times the information Theory approaches deal with information geometry so which one is it I agree that it's enough to call it mathematical and just say includes topology information Theory thermodynamics geometry just a lot of math stuff um another area of mathematical Frameworks for Consciousness could be like cybernetics which is why it was one of the keywords for this paper because a mathematical framework for Consciousness might involve decision-making or agency or ability to plan at a certain level of coherence and in fact in the 2018 interview that was with Carl friston myself and our pass away colleague Martin forer in the last question of that interview I asked him whether he thought the ant colony was conscious because I was working on this paper with Eric and I was writing the interview with Carl and then I just thought wow we could figure out what Carl thinks about ant colony's being conscious and if you're interested in either fris andology or in ant colony Consciousness then definitely review Carl's answer on that but just to kind of close this slide off um we're committed to long-term Clarity on these issues about the study of the science of Consciousness so we really welcome collaboration or Improvement on these questions let's work with it let's make an ontology for Consciousness or how are we going to get organized on this issue because as I said there's just too much on the table to let this one slide and we have to enter this space and do it right because otherwise just it goes off the rails okay let's get to neurophenomenology which is one of the keywords of the paper oh yeah the paper and on the bottom here I have a couple just fun images the left side is the book GB by Douglas hoffstead 1979 but reads as crisply as ever and uh really an insightful book and then on the other side uh of uh is the tree of knowledge this yellow cover by verela and matara and that's a book about autop poesis and a few other topics that are really cool related to complexity and just very deeply related to the ecological an activist semantic biobehavioral lot of great stuff also potentially the um work of frof cpra is one thing to look into here but then some images because it's not all about language so here are some images by mcre and by mcer um that have to do with sort of what is this self let's look at how the authors Define neurophenomenology they Define it as one observation and description of the phenomenological invariance at the appropriate level of abstraction so maybe the appropriate level of abstraction um for the experience of drinking water is not uh you know a throat cell maybe it's uh not a whole nation maybe it's uh a person the classification to classification of the behavioral cognitive and biofunctional aspects of Consciousness so some drugs like you know you take acetamin and it doesn't make you feel that different there's other drugs that are psychoactive or that have long-term changes to your brain whether you experience it in the short term or not so how are we going to get a behavioral cognitive molecular genomic understanding of that mechanism three identification of the neuroanatomical and neurophysiological correlates of Consciousness so I see neuro Anatomy as the structural whether it's F uh fun structural at the macro or the micro neuro Anatomy is what sovic and I called structural then neurophysiological is um referring to the Dynamics and the processes molecular informational whatever they may be that are in the brain so those are the functionalist models to me of the brain four development of mathematical and computational models of Consciousness which I would consider to be in the mathematical category that a Faithfully integrate all the invariants described in one which we'll get to soon B provide unifying explanations of inestable hypotheses about the biofunctional aspects of Consciousness described in two all on board and C can be physically implemented in the hardware identified in three as well as perhaps in non-biological Hardware so I don't subscribe to computationalism uh I don't believe in a hardware software distinction except as a useful metaphor in biology I think the book wetwear is pretty interesting on this topic and the reason why I intervened right after the mathematical and computational models is because I think it's actually a different domain I think there could be a mathematical model that's extremely agnostic to the biology or our biological model could just be like it is what it is and it doesn't matter what the m or what anyone thinks the math is it is what it is because of evolution or um because it passed through some sort of historical filter and any equation you put on it is just like a linear regression it just it doesn't matter so those to me they they can kind of fly together and that would be great or not it could be something totally off base which is why I think again it's important that even though we have the citations and the ways that we can sort citations into keyword clusters let's not think that we've mapped out the space of the possible in consciousness because there's people in other cultures who don't feel the same way as scientists first person perspective is I think the last keyword and they write about that in the paper they say ideally this method of the PCM would give us a theory of Consciousness that intelligibly links the phenomenology of a firstperson conscious experience with its realization in the brain or other substrate via mathematical and computational model simultaneously accounting for its biofunctional properties so the B functional properties is kind of where you got the um neuro part the mathematical computational model they're saying wouldn't it be good for purposes of describing it predicting it controlling it designing it um explaining variants reducing our uncertainty why not that's the mathematical computational side and then pull it all the way back phenomenology so computational neurop phenomenology we're going to be doing mathematical computational neurop phenomenology in this paper and we're going to be using geomet tools like projection mapping to think about how cybernetics and active inference are related to Classic work in neuro phenomenology okay one more um key word perspectival imagination so they quote on this one they write this perspectival space of the field of Consciousness FOC is organized around an elusive implicit origin or zero point and includes a horizon with Vanishing points at Infinity that Mark its limit so right there there's the two of the violations of ukan geometry point of privilege that's the point of view and then the vanishing Horizon points at Infinity the FOC embeds vectorized directions that frame the orientation of attention and action along three axes vertical horizontal and sagittal so like you know this one this one and then that one these are given in perspective with respect to these same points at Infinity so pretty interesting stuff they continue and they write since we are dealing with a 3D perspec hial space with an elusive origin point of privilege point of view and Vanishing points at Infinity constituting a horizontal plane a horizon plane that happens to be in the horizontal axis doesn't have to be if you're on your side it's not the PCM postulates that the field of Consciousness has the structure of a 3D projective space in the sense of projective geometry nice thanks for giving it as a keyword too it helped me learn about it the FOC can thus partly be understood in terms of a 3G projective frame let's look at a few just kind of cool examples related to perspectival imagination so this is an example of a non- ukian geometry it's an eer uh artwork and it's like these bats that kind of fractal off and sometimes this could be related to hyperbolic geometry or other geometries but the point is like each bat might be flying off and so that kind of maps onto our experience of a bird or a bat flying away from us and slowing down perhaps um but how is that going to get mapped onto another space that may or may not have three dimensions spatially again Bucky Fuller might claim there are four spatial Dimensions so let's not assume there are three spatial dimensions and that the fourth one is time that being said it makes sense to talk about XYZ coordinates from a cartisian perspective another example about this perspectival imagination is vanishing point so everyone who's done a fun drawing course or just uh looked down a city road or looked out of a car that was driving through Fields everything kind of vanishes and so you can see the rows of a field and you know that they're all in parallel but then you can see really deeply down one other ones like kind of move towards it and another way of seeing it in a cityscape is like you can conceptually know that the street is straight but if you look at it it looks like a triangle that's disappearing from you with a point at the top so what is it that allows us to be like oh yeah buildings have 90° angles and they're going upright even though all the actual angles are kind of whack so that's this perspectival imagination and that's what makes an artist who can make a lifelike drawing especially like an architecture or something look lifelike because they capture these extremely subtle features about vanishing point and perspective and all these different things so one drawing that I love on this is um this one and it's like for many people it's said okay do a self-portrait of what you see when you um are on the couch and one approach might be like I just you know draw like okay I see this you know here's this piece of art here's my bookshelf here's you know my bike then the next layer back like oh I'm going to include my feet in the drawing like the kind of you know I'm on the beach here's my legs type thing like uh taking a picture from right in front of you with a phone something that we actually have an affordance to do now but other people may not have seen in such a direct way um which is why they didn't take selfies for example oh hey here I am this drawing takes it back another layer now this is actually where are you really seeing from not what is canceled out with a binocular vision but what happens when you look with one eye really really far to the side well you see this curving and it's just like a little bit beautifully accentuated but of course we're seeing that as a two-dimensional projection so that's just so cool and uh styles that play with our perception like optical illusions or cubism or abstract art they often incorporate some invariant features of perspectival imagination but then violate expectations on other ones so like um half of the person's body is as if they were lit from this side and half is if is as if they were lit from a different side that was a classic technique of the cubists and so very rich space very open for honest collaboration with artists and philosophers and practitioners so we hope that this is kind of an opening for those who might have thought active inference okay it's an engineering framework we're talking about optimization we're talking about sociocultural things but kind of from an abstract point of view how are we going to take this into our experience of Consciousness and of perspective so I'm going to return to this slide again and again which is that it's a question what does free energy principle and active inference say about the relationship between the world and agents so that's this feedback between the world and the agents and the nested agents and here we're adding on this level of the agent is having conscious awareness it's thinking I am a strange action perception Loop or at least it's acting like it says it or it's thinking that it's saying it and just as we've talked about before with this slide we want to be building on the fields of action of inference of physics of experience we want want to be tying these threads together while also being aware that just being convinced will never be enough so that's why it's such a fun area why it's so great to do research in this area and to learn about it and also to welcome collaboration and to welcome people sharing their perspective because I really believe that no matter how many papers you've read about Consciousness it does not make that opinion any more valid there's other topics where potentially you couldn't say that as flippantly but in the area of Consciousness it can't be a counting game it's just not it's something that is so essential to our experience and so it's such a deep area a perfect area for transdisciplinary synthesis cool let's get to the paper because it's also all about the specifics we want to be really specific about what the authors say and do so that we can build off of it with other specific work so that we can actually um use their perspectives as catalysts for our own development not push ideas out of our head and replace it with you know the frisen update but rather be catalyzed and spurred to new culturally informed creative states by reading and by discussing so that's why really especially for 9.1 and point2 for the last home stretch of 2020 and for 2021 just if you're listening to it we want to know your experience and we want to hear your questions in the YouTube chat or have you on live however it has to be this paper is called the projective selfhood it was published in December of 2018 in the uh Frontiers and psychology journal in the division of theoretical philosophical psychology and the authors are Will aord benoquin friston and redr so that's kind of giving you some information on when this paper was published and on what areas the authors want to place their work within let's take some quotes from the paper and look at the main aims and claims of the work they write can we describe this integrated multimodal and centered spatial structure of conscious experience in a rigorous way great question I love the setup and they WR and can we give a good account of why it should be thus organized here we answer both questions in the affirmative great questions I love the setup and it's something we can evaluate whether that's a hearty affirmative or a half-hearted or a overly confident or underly confident or okay it's affirmative but then what do we do these are all great questions to have and I just love that the authors can be really clear about this great research and also a great suggestion by a Community member to um read this paper overall the PCM delivers an account of the phenomenologically available generic structure of Consciousness and shows how Consciousness allows organisms to integrate multimodal sensory information memory and emotion in order to control Behavior enhance resilience optimize preference satisfaction and minimize predictive error in an efficient manner sounds chill if we could we should great the way that I would rephrase it without um any of the uh jargon of actm is how can we apply active inference to the study of Consciousness in all of its unified yet coherent mysterious Dynamic and profound facets so there for the purposes of scientific writing highlighted a lot of the vocabulary keywords and research Fields related to control theory information signaling geometry math computation those are some heavy-hitting ideas but let's also not lose sight that some of the other heavy-hitting ideas include the mystery of Consciousness and why we're the one that we are and whether cells have Consciousness and whether ant colonies do and whether colonies do but not nestmates and nestmates but not colonies or ecosystems but not colonies just let's be open-minded so let's answer questions in the affirmative and be open-minded let's have yes and and let's accomplish yes and with everyone's participation the abstract begins with saying we summarize our recently introduced projective Consciousness model and relate it to outstanding conceptual issues in the theory of Consciousness the PCM combines a projective geometrical model of the perspectival phenomenological structure of the field of Consciousness with the variational free energy minimization model of active inference yielding an account of the cybernetic function of Consciousness viz the modulation of The Field's cognitive and affective Dynamics for the effective control of embodied agents long sentence double double points for um using uh affect and eect uh long sentences but slow it down unpack it I think that we've gone through the keywords well enough at least a first pass to show you why it would be great to connect across all these areas abstract part two the geometrical and active inference components are linked via the concept of projective transformation which is crucial to understanding how conscious organisms integrate perception emotion memory reasoning and perspectival Imagination in order to control Behavior enhance resilience and optimize preference satisfaction so sounds cool the PCM makes substantiative empirical predictions we'll see about that and fits well into a neuroc computationalist framework of course it does it also helps us to account for aspects of subjective character that are sometimes ignored or conflated pre-reflective self-consciousness the firstperson point of view the sense of mindness or ownership and social self-consciousness and close the abstract with saying we argue that the PCM though still in development offers us the most complete Theory today of what Thomas metzinger has called phenomenal selfhood now I don't know meter's work incredibly well or what this topic Bridges out to kind of an interesting way to end the abstract but so it goes um that's what we might be getting to how are the authors going to get from all of these keywords like cybernetics and geometry to getting to potentially the most developed framework to date in their opinion of what is called phenomenal selfhood so phenomenology experience selfhood having a self well they start in section one with the introduction and go pretty directly to section two methodology I think this is great because it really gets into the details quickly you could philosophize for a thousand books people have so what's the methodology how can I identify what you're actually doing in this paper as a researcher um without hearing 30 pages about your understanding of what DART did or didn't say three they're going to talk about phenomenological invariance and the functional features of Consciousness which are also we're going to go into them later and they talk about the phenomenal logical invariance what they are five of them we'll go through them and then talk about how they're related to projective geometry in the PCM so basically they're going to start with their invariance and if you have a qual with an invariant you want a sixth one you want to collapse to four great let's write another paper but they're going to start with these invariant features and then they're going to introduce that plus projective geometry to say projective geometry is the bridge we're going to take to get these five invariant features we see figure one which has an account in the PCM model of psychological phenomena and figure two which shows projective geometry in relationship to the Necker Cube which is like kind of a cool illusion that I'm looking forward to showing you then they go to the five functional features of Consciousness which are also what we're going to address and they talk about the projective Consciousness model which is remember the phen phenomenological invariance plus projective geometry equals part of the PCM now we're going to talk about the PCM and the functional features of Consciousness the functional cybernetic functions that they talked about earlier and fuse that in the capacity of using active inference and free energy minimization to get some of these functional Dynamics then they'll show figure three a simplified 2D um projective Consciousness model based simulation of a pit Challenge and then figure four give a overall sketch of the projective Consciousness model in section four they talk about the hard problem of Consciousness representationalism and phenomenal selfhood the hard problem and computational functionalism is um variously phrased by uh professor chers and others as basically being about what the substrate of Consciousness is why is it that when you hit your hand it hurts but when you hit some brain regions uh it doesn't hurt and when you hit some other brain regions you just pass out you come back or you die or something in between how is that happening in the brain or is it an antenna is it a generator these are the questions then they go from the computational functionalism to talking about the PCM and representationalism now all these terms computation function and representation are pretty loaded we've gone into them on a couple of other discussions but be open-minded as to how they could recombine in the context of PCM and then they're going to take those kind of more quantitative terms computation function representation representation is a little bit of a bridge and they're going to bring that into the projective Consciousness models phenomenological components so that's going to be about subjective character subjective character being what it isness the what is it about to be about and also the the phenomenal selfhood of the philosopher that they concluded with discussing in their abstract and then they conclude so looks like a great road map let's in this introductory video in the um time that I've left here let's go through the functional features and the invariances of Consciousness because we're not going to have time to go through them in 9.1 or 9.2 with all of you contributing your perspectives and it's good to get them out here because these are kind of like the big five for the functions and the big five for the invariances and if you have a qualum with the model you should first ask is there a function or is there an invariant that I think is extraneous um is there one that I think is missing is there one that I is just off base so if you have a question about one of these five things you think this doesn't go far enough or this goes too far or but some conscious systems don't have that great that's the question you want to ask so there's five in each category if you think there should be more or less or a different one that's one entry point now if you agree with all them and you say yep those are the perfect five in each category then you can evaluate given what projective geometry is and what the PCM is do they May address the functional and the um uh experiential invariance of Consciousness so that's how we're going to read this paper and that's where I hope there's a lot of footholds for people who maybe have thought about Consciousness but this is a new way to hear about it for them it would be great to have their comments what are the five functional features of Consciousness according to the authors and the way that they sort of justify this is by saying um we claim that the overall function of Consciousness is to address a general cybernetic problem or problem of control control theory Consciousness enables a situated organism with multiple sensory channels to navigate its environment and satisfy its biological and derived needs efficiently this entails minimizing predictively erroneous representations of the world and maximizing preference satisfaction a good model of Consciousness must be able to explain how this is accomplished the PCM thus emphasizes the following interrelated here's why I pause before putting it up but after just in it because even things that do control theory might not have Consciousness full stop that's the whole thing that I was getting into with the ant colony test is okay so then if I have a robot and it has something that allows it to do multimodal integration across sensors with a generative model with a deep with a counterfactual whatever it is does it have Consciousness and then that's where the person goes oh well it needs a brain and so then they go down the rabbit hole a different way so the function is important but again again it's not the whole story especially yet here are what they characterize as the five functional cybernetic features of Consciousness that we'd want to at least explain so maybe just passing these five is necessary but it's not sufficient but it's important one Global optimization and resilience Consciousness helps the organism reliably find the globally Optimal Solutions available to it given competing preferences knowledge based capacities and context making the organism more resilient this is a mechanism of global or subjective here in the of subjective basian optimization which does not always obil objective optimality though it must approximate the ladder reliably enough uh or you die if you don't objectively enough figure out how to walk down a staircase you're not going to walk down many staircases now a few things on this first many people think Consciousness is epif phenomenal so whatever is happening in our awareness is not going back to guide so this plays a little fast and loose because it's saying well Consciousness helps your organism reliably find it sounds like Consciousness is is coming back and this is the whole cartisian dualism Paradigm is like I know that Carl friston and others have many more nuanced Works including I think therefore I am related works so I'm not going to go into it but it's like if Consciousness is something that reaches back into the physical how is that or if Consciousness is purely a descriptive epip phenomenal epip phenomena above the phenomena of mechanistically or um materially defined components like neurons message passing whatever it has to be ants no no matter how emergent their properties are in the weak emergent sense do they ever amount to strong emergence and actually cause the system even the um downward causation and causal entropy and the renormalization group that still is a very philosophically agnostic with respect to whether it's causal so another thing to say here is that Global optimization given the preferences knowledge based capacities affordances and context so so it's not Global optimization so it's local optimization or its Global exploration of a conditional space conditional upon the things that we want it to be conditional on like how far you can reach but then I don't know about you all but my Consciousness doesn't always help me get to the globally optimal points unless that Global Optimum is defined as this not objectively best space of like well you you just didn't go the right way but it seemed optimal given what you knew it's like okay and then okay and it has to be reliable enough so I'm on board two Global availability the contents of Consciousness must be generally globally available in the sense of poised to be treated of by our effective cognitive and perceptual systems as the occasion requires so for example you know here's idea 1 idea 2 memory 4 the color you know the number all these things are just Apparently one layer away from accessing the workspace this is drawing heavily on the global workspace Theory attention schema model of some other researchers I would just point out Global doesn't make sense to me the brain isn't the globe cognition isn't the globe availability it's like yes but then again you define available as like I knew it yesterday but I don't know it now so it's not available um or what if it's part of my extended cognitive process and it just takes me five minutes to find it so okay relevant but globally available um and also our awareness is hardly the tip of the iceberg um or maybe it's a whole Iceberg and a half so I wouldn't even know what it would mean to be globally available motivation of action and modulation of attention attention schema Theory Consciousness is generally implicated in the motivation of action based upon affective Dynamics and memory this drives the modulation of attention though Consciousness is not to be narrowly identified with attention so that's kind of attention schema Theory Breaking Upwards out of the just um like uh lstm like a neural network that has an intentional parameter we want to go into the experiential Dimension not just the oh well it's a laser that spends a lot of its time here or um you know it's a camera that autofocuses on this level we want to go beyond that and say um whether or not it is getting input like a camera in some of its sensory organs the experience is somehow generated but I'd also point to this motivation of action and modulation of attention it's very very important these are phrased as if Consciousness functionally functions of Consciousness right motivates action and modulates attention or it could be epiphenomenal Consciousness that is and so it could be the experience of motivation of action and the experience of modulated attention so in that framing we made Consciousness epiphenomenal and we basically said yeah attention is something that happens with all the neurons and the firing patterns and whatnot you know the drugs everything that happens and then it gets to some synergetic compromise where it's like looking at this thing at this time and then you're just experiencing it it's not simply a cartisian theater model it's still a bottom up top down um situation but there's no simple outs here it's neither that Consciousness is reaching back and motivating action neither is it that it's modulating attention through what dials you know show them to me oh well that's what meditation is yeah but then all of a sudden you're getting into this area where I think a lot of stuff that's not related is pretty important and then the fourth one oh whoops I said big five well let's just think of it as the big five the big fifth function is going to be the secret function but the fourth function is simulation enhancement maybe that get us too Consciousness facilitates the use of simulation in the broad sense of imagination not the narrow sense like simulation Theory um but maybe that's a direction to go in the service of solving cognitive perceptual and affec of problems and more generally in the anticipation and programming of actions and the response to their outcomes so simulation enhancement could be for example um I wake up and it's very dark in my room but I have a generative model um or an accessible model not um being too formal here of my room so even with my eyes closed or with very dim light I can enhance my simulation and think oh wait you know if I open the door this way it's going to wake up my roommate or if I do this at this time then down the road this is going to be a different sensory input that I don't want to experience so that kind of Rich U mechanistic understanding of the world is what humans excel at and it's what the current machines that we deal with do not deal with well um you can ask it a question like could you know this building wear ice skates and it might be able to parse whether um those terms mean different things and whether they're keywords used by the same people or not on Twitter but will it be able to understand and address that question and that's kind of where the GB book really is fun to think about so again four functional features for Consciousness should there be a fifth one are one of these extraneous do you have a different function for consciousness what is your understanding is consciousness have function or is consciousness something else how would it affect things if it had a function how would we measure the function what is dysfunction all these questions any questions that people want to have maybe be cool for people who watch it like to post questions they have and we can have a norm of just asking the questions that we have out loud because let's ask the questions we have I don't know any other way to say it right now the second thing that will we'll talk about is the um invariance of Consciousness the phenomenological invariance and here I believe there are five so these are the things that are really what their model is going to come down to with respect to phenomenology and philosophy so this is what they're going to focus on this is what the mathematical model needs to aim for this is what their formalisms should explain predict integrate Encompass um so this is like the big core philosophy idea that we want to do all of the math and all of the computational stuff to get at with some type of bridge the bridge is projective geometry perhaps so what does it mean to be a phenomenological invariance and they write the following are the deeply interconnected invariant that we think characterize Consciousness we take them as postulates for mathematical treatment citation that you could follow up on and construct the PCM on their basis so it's kind of like axioms like they're taking these as like roots statements that kind of don't need to be justified they're kind of like observations but of course this is something that we all want to know is could there be another one could one of these be redundant what if you have four out of five do you get 80% on Consciousness or what if we figure out or Imagine A system that has uh three out of five is it you know what does that mean for that system what are these phenomenological invariants one relational phenomenal intentionality so break it down by the words all Consciousness involves the appearance of a world of objects properties Etc in various qualitative or representational worlds representational ways to an organism so again they're not thinking too much about worms and fruit flies and ants and ant colonies maybe they are I'd love to hear it from an author or from a colleague if they are but um these are the debates that we're having which is like right if You observe the worm moving away from the gradient as if it had a representational relational phenomenal intentionality is that evidence enough and um that's the question two situated three-dimensional spatiality of course also there's time sometimes called the fourth dimension the space of the presented world like objects or Etc is three-dimensional again big asterisk and perspectival unfolding in an oriented manner between a point of view and a horizon at Infinity we all parallel lines converge the origin of this point of view is elusive though normally it seems to be located in the head the space is normally organized around the lived body so this is like bringing us to embodiment to in activism to ecological psychology to Intercultural and developmental psychology to um Ken Wilbur's integral Theory also connecting us to lived experience and to spatiality and to the ways that the spaces that we live in influence who and how we are so all of the awesome ACM stream participants who have taught us so much about embedded systems and embodiment we'd really love to hear what is work in this area that impinges on one of these invariants multimodal synchronic integration synchronic means at the same time Consciousness involves the synthesis or integration into a unified hole of a multiplicity of qualitative and representational components from sensory modalities memory and cognition so dionic means at a different time synchronic at the same time so memory is dionic but it's of sensory modalities so it could be a hearing event from six months ago um maybe you know and there's the whole question of imagination obviously big question but just to a first pass um events from the past that influence us in the future are dionic or another way to think about it is they modify the system or they put it onto a different trajectory that's different than if that differen making cause to use the waters terminology hadn't occurred Consciousness seems to be both dionic and synchronic like when we're listening to music and seeing flashing lights there's like synchronization in the same moment that's also multi-sensory but there's some really interesting work about the time durations of different kinds of experience so clearly something is happening there with Sight and Sound and just people who are born with or without hearing and born with without sight all these different people can navigate in the same world so that says something about sight you know it's not just us watching a projector if there's people who can navigate without sight so that's kind of what they want to bring it back to four temporal integration which I just hinted at before but this is a little deeper level Consciousness involves at least the retention of immediately past experiences and pretention of immediately future experiences as integral elements of its specious present the uh everpresent now as it were and a foundation for more expansive forms of temporal ation such as Distant Memories and long-term planning distant past distant future now just to give one kind of funny note when I was reading this I thought what is protention what does it mean in this context so according to Google protention is a philosophy word that means anticipation of a future event so I dug a little deeper into the Mariam Webster definition of prot tent P R O T N T Pro tent like a professionally setup tent it's an archaic transitive verb meaning to stretch forth or to extend now nowadays one might hear about portending future events not penting them um and the words actually have different meanings so to portend p o r t n d portend is to um convey something through like a symbol or an omen like the the owl portended difficult times or something like that but um port tenting is something that the agent does in response to the symbol so um upon seeing the owl you know the farmer decide pented that this was going to occur and so I would just kind of bring up this idea that maybe they're so similar sounding and they're so kind of close in their word usage that the English language just couldn't tolerate um that much ambiguity because penting and portending they sound like 99% identical and so then maybe portending took popularity over prent because it removes agency from the actor and so it's one way to remove action from language and so now proten or penting it's now a philosophy word it's archaic it's an old word because we can't have action in speech right no but non-ironically it is really interesting to think about how these words change in meaning and the more action we can embed in our speech patterns and our interaction patterns the better subjective character um so subjective characters defined by the authors as involving a pre-reflective non-conceptual awareness of oneself and their individuality now one could be totally off base so they could be wrong is that part of Consciousness to be wrong about your individuality or is it just that you're having an individuality out uh hard to say but the authors have it as an invariant they write also that two to five are in fact implicated in one so relational phenomenal intentionality is really the most important key invariant so that's the relational thinking and that's why I feel like the active inference and free energy principle communities are very adjacent to complexity thinking to systems thinking relational thinking design thinking as well as um team collaboration Frameworks because so many of these social work and team relation Frameworks as well as other types of thinking including complexity they all deal with this relational Insight so there's so much to learn and to do here and to understand about what are the implications of relational thinking on Consciousness on our own experience if not on actually making claims about Consciousness cool let's just kind of close out by going through and looking briefly at a few of the figures and just what they're about take a 30,000 foot view so that when you're reading through the paper you kind of know where you're coming from so figure one has um a or and c and and um a is a projective solution to a VR VR mediated disassociation in phenomenal selfhood so this is kind of like um from the experience of uh the ukian view like we're looking at Minecraft here and then the person their head is in this like Block in the lived space in the right side of figure a and um but they're Imagining the space as if it were on the left side of a so I can imagine my office right now now in terms of the right angles again even when I look out I don't see right angles if I were to combine like this and I look back down and I look at the angle that the room was just making it's a not 90° angle but it looks normal within my generative and usually verified experience that it is 90° because why projective geometry so what do we have to do well it turns out to add in this element of the visual field in the projective geometry in B we're going to show this like sort of they call it an anti-space beyond the plane I'm not even going to go into what that may or may not be because I don't want to um go down too deep but it's kind of like there's some mapping reflected in B and C what they call God's eye Vantage Point there's certain mappings from B and C just like there's a mapping between the understanding that I'm in a ukian office and then my lived experience where apparently wherever my eyes are or something like that that right a little bit behind there a little in front of there um how are we going to square the circle almost literally with the generative model of the ukian office with my lived experience of the wacky angles and what's interesting here is the visual focus and so where are all these different centers is this kind of related to like what if we feel like our emotional Center is somewhere else in our body or what if we um have the experience of not just a simple out of body like floating above your body looking down but more expansive experiences so would love to go into that phenomenology with someone potentially from Alias or any other colleague um and so there's certain mappings just like ukian to non- ukian office there's this like God's eye mapping you can evaluate for yourself whether you think that's God's eye figure two is a projective solution to the Necker Cube and so here's what's happening in this image on the left side is like two versions of the so-called necro Cube illusion so it's a cube visual illusion and it turns out it's illusion um first off just looking at these two for one or for both just ask yourself do I see them as um how should I even say this the biggest side that's close to you is it on the top left of each square or is it on the bottom right and you can flip some times some people different people in different ways can flip in their mind seeing it as if the biggest side of the Box were coming down to the right or whether the biggest part of the box is kind of going up into the left okay now here is why they are suggesting that this um visual illusion exists because there's no perspectival cues we're looking into a projection on a flat plane and so there's sort of like two Optimal Solutions to the perspectival question and so because there's two such similar um evidence competing models free energy minimizing models it's B it's not they they it's undecidable free energy based inference which is to say that in our optimization perception we continue just oscillating it's like you know it's the old woman it's the young woman it's the old woman it's the young woman okay that's a visual illusion but this is one that has to do with projective geometry where it's like it's coming towards me like but I'm getting sensory evidence it's going that way in a different simulated frame and then you perceive it that way and now it's top down so it's like this um back and forth resonance and there's a lot of interesting research on this what breaks the Symmetry quite literally and the answer is lived experience from the point of view simulation so here if I'm closer to one side of the square I don't have any little squares to pick up but it resolves it not absolutely you still know that square has a different side or the cube has a different side and you can even imagine what it might look like like oh I'm seeing the six is on the die it's closest to me if I were on the other side I would see a one and it would look bigger and I wouldn't be able to see the six but where I'm at I just see a big rectangle six or something like that and so that kind of captures a lot of these aspects of the geometry of Consciousness where you can simulate being in different places but you're always in a certain SP spot visualizing it what is that spot that's the privilege point of view that's the point of reference what about it is special well it's like a non-arbitrary point in a projective geometry what kind of projective geometry well something that takes our allegedly ukian world and Maps it into this like vanishing point experience I think I got some of the main uh threads to link there but that's kind of what they're suggesting and free energy is going to be the optimization and action and inference Paradigm within a simulation broadly construed understanding about how symmetry is broken in terms of experience so big stuff interesting topics you know to be wanting to address this is big by the authors figure three is a uh PCM based simulation of a pit Challenge and on the very bottom you'll see this is like from another paper so if you're curious about this simulation definitely go check out the other paper that they reference but what we can look at here is um starting on the left side so there's like a bunch of barbell shapes and basically the barbell uh has to do with it's like a map they can walk around in so they can wander around but they kind of have to make it across this pit to get to like a rewarding place um so there's an anhedonic in a safe space a challenge room and a goal room I believe is what they call them and then they're going to be thinking about that that moving circle like let's see if I can do this there we go this moving circle is like the agent moving through the map here's its expectations and where its prior beliefs are and then here is it resolving its uncertainty in a it's kind of like looks like a radar dial but that white spot in the middle is like the the nonarbitrary origin that's like the Consciousness Zone and so from the POV point of view of this organism in its FOC its field of Consciousness which is like some geometry mapping from the um experienced space into its generative model it's navigating that mapping somehow through Consciousness and then that's also able to bear on the different between like having a weaker imagination and a stronger imagination they have this active inference Engine with um multi sensory integration and this looks like something definitely to go into for the 9.1 and for our own future uh research but basically we can run model scenarios we can say yeah I guess if it it's a funny thing to say you know I can imagine that if you were to do this experiment you might find inform ative results suggesting that enabling an agent to have imagination in an action oriented geometry might enable it to do better than a model that doesn't therefore spending most of its time learning about relationships that may or may not be relevant in the geometrical frame that sensory data actually occur in again long sentences abound but I think that um there's a lot there in figure three and also a lot to learn about how they're framing free energy all right last figure figure four figure four is an overall sketch of the PCM the projective Consciousness model and I think uh there's a few different dimensions here so one is the Observer at the center with their lived space uh Stephen this really reminds me of the Parry personal space so I hope that you'll have a lot to teach us in this domain and within this domain this field of Consciousness field of affordances let's think about it from many different perspectives just by saying one thing I'm not always meaning to limit I'm just saying that this is one way we can think about it let's imagine that there are perceived and imagined events now observations are just models of observations so maybe there's less of a distinction there than might be seen but in this geometry Observer Centric with all of this projective geometry mapping happening can we think about aligning your vector through time down a gradient that goes from red to Green so it's funny because it's a gradient that colorblind people of certain types may not be able to perceive which just so perfectly entails everything we're talking about here which is that um action in some like absolutist sense is moving downhill thermodynamically there's very few people who would say that action moves uphill thermodynamically bring them on the Stream but it has to go uphill conceptually or symbolically in certain cases if not just simply logistically in terms of going uphill to go downhill later and so here we see a gradient reflecting the free energy going um from a culturally based green is okay and low surprise and then red is high surprise high alert high attention where have we seen that before um taking that kind of a culturally imbued color scale and then aligning a Time Vector as a dimensional projection onto that figure as a way to say that actually through space and time in the context of the projected geometry and affordance capacities of the organism it is still downhill in a sense so I don't know where this um necessarily fits in with uh all the mathematical formalisms here looks pretty nice to talk about like the action perception prediction imagination Loop this is another loop we might want to learn about we've talked a lot about loops related to like agent and environment but now we're almost thinking like the agent within the agent is going from action perception prediction imagination so that's something happening inside of us potentially and then um again on the right side is looking at time series that's the directions we want to take all free energy research is like through time through space any other dimensions we come across we'll go through those too but that's definitely the cool Dimension to think about is deep causal inference with the sound and what's visualized um in the figure so other people may see very different things in this figure and we'd welcome hearing about it let's close with some just um final thoughts and questions to end on which maybe we'll start ending with questions more they write in their paper computationalist a poori that means after the fact identity Theory allows that apparently contingent processing constraints may be essential to real Consciousness as long as the computational model captures the phenomenological invariance meets the functional constraints and survives the tribunal of prediction and experiment we should happily accept that some apparently contingent and only empirically knowable constraints pertaining to the physical realization of Consciousness are actually essential to it so that is basically like saying look if we do the Phil if we can explain and predict the invariance the five invariant and we can capture these functional attributes there's four of them but again both of those sets the invariances and the functional attributes anyone is game to add remove or change it just needs to be justified and understood so they're saying look if it does both of those things which is why I spent so much time on it in this talk was because that's their claim it's not just broadly we could do active inference and think about Consciousness it's saying if we make these invariances part of our model and we at least get all the functional attributes that we want then any model that is surviving unique explanation and predictions the tribunal the uh the tra as it were if it does all of that then it is basically going to come along with some wacky baggage like a parameter we don't understand or the God's eyee view and then they're just saying you're just going to have to accept that as contingent so fun thought by the authors just a really well written and conveyed paper and I hope that it broaches onto many areas that we've talked about in other actm streams here's my closing set of questions one how would we recognize a framework for Consciousness and uh a related question is what might a framework for Consciousness be what if it's a poem Oh Well it can't be a poem how do we know or what if it's a drawing or what if it's a set of ontologies narratives formal documents and tools could it be that on nft could it be bolts business operational legal technical social um could it be acceptance could it be fundamental agnosticism um could it be free energy principle how will we know and that is really the key question for theories of Consciousness is not do you feel right it's how will we know and do you hold yourself to that standard because we can do a lot better in this area and this paper is an example of great work in the space our next question is what might a framework for Consciousness enable and that's pretty broad what would it change in our action practice would we put more people in fmri Scanners or less would we communicate with people who we don't think of as being able to communicate with at this point or not why not um how might it change our legal systems or our uh social systems what if we could measure different kinds or types or levels of consciousness and then that's really why we need to get down to the Brass tax on what it might look like what it measures and what the predictions are because if we don't know the unique predictions or experiments and this is alluding to earlier when I said that there's a lot on the table someone's going to say well I have a Consciousness Omer and this is a two and this is an 11 and um this type of object that you care so much about is actually it's not conscious so I'm just going to unplug it but this other one that I care a lot about I measured it with my device and it was very conscious it's very worthwhile to save and where will people make a stand and the stand has to be on the common ground and the space of recognition of Frameworks for Consciousness in my opinion I'm open to changing my mind on that but I think if we don't meet at how we'll recognize such a framework what it might look like and start with respecting World Knowledge traditions and by respecting differences in our opinion and we stay in a hegemonic mode in a um absolutist mode in a um non pluralist frame culturally methodologically if we don't go down um the pluralistic route it's going to be I think very bad for practical implications and then lastly um what are the next steps for active inference in this domain what do we want in active inference conscious agent to do or how can we build on active inference now do we want to go deeper into Consciousness deeper into action um we spoke about action policy selection and machine learning could we put Consciousness in machine learning what would be the philosophy and the implications of that these are all big questions but how are we going to take this paper and develop in the directions of active inference even if only making um educational content and then lastly um maybe always a good question to address is what is the goal of This research sure we're curious about Consciousness but aren't we curious about so many things what makes it worthwhile to do this research why um are we doing it how have we selected our how why have we selected our how how have we selected our why every one of these combinations we really need to keep in mind especially if it's about us about our experience about Consciousness about people it's just again can't be said enough that there's too much on the table to not have this be participatory inclusive crosscutting Intercultural the list just goes on so everybody who's listening and thinking about it thanks so much this kind of uh I guess went a little bit longer than one might have predicted but say right when active streaming why not thanks for participating we will provide follow-up forms to the live participant we actually will believe it or not we will um request your feedback suggestions and questions so if you have any thoughts again I tried to drop a bunch of times really specific moments where somebody could just pause the video and say yes this thing right here I do or don't agree with or this does or doesn't resonate with my experience that would be epic because let's hear from a lot of perspectives on Consciousness it's important to hear about that and then also just any feedbacks or suggestions or collaboration offers for the ACT INF stream always welcome you can stay in communication with us any of our channels or means but thank you all for listening those of you who were listening live or those of you who will listen in replay because the future is always Now isn't it see you later